

WINDOWS PROCESS CREATION INVESTIGATION: EVENT ID 4688

1. Title

Investigating Windows Process Creation Using Event ID 4688

2. Objective

This investigation examines Windows Security Event ID 4688 (*A new process has been created*) to understand how process creation activity is logged and how it can be used to reconstruct user and system behavior on an endpoint.

The objective is to:

- a. Understand what Event ID 4688 records at the process, user, and session level
- b. Identify the creator (parent) process and new(child) process relationship for each logged event
- c. Analyze the command-line arguments associated with process execution, where auditing permits
- d. Correlate Event ID 4688 activity with the corresponding Event ID 4624 logon event to tie process execution back to a specific authenticated session

3. Investigation Question

“What can Windows Event ID 4688 reveal about processes executed on a Windows endpoint, and how can an analyst use process creation events to reconstruct user and process activity?”

4. Environment

<u>Component</u>	<u>Detail</u>
Operating System	Windows 10 Virtual Machine
Virtualization Platform	VMWare Workstation
Log Source	Windows Event Viewer- Security Log
Primary Event	Event ID 4688
Correlation Event	Event ID 4624

5. Event ID 4688 Overview

Event ID 4688 is generated by the Windows Security Auditing subsystem whenever a new process is created on the system. It is logged to the **Security** event log and is one of the most valuable events for endpoint visibility in DFIR and SOC investigations, because it provides a direct, timestamped record of process execution.

Each 4688 event typically includes:

- The Subject (the account and logon session that requested the new process)
- The New **Process** Name and New **Process ID (PID)**
- The **Creator Process Name** and **Creator Process ID** (the parent process that spawned the new process)
- The Process Command Line, if command-line auditing is enabled
- The **Token Elevation Type**, indicating whether the process ran with elevated (administrator) privileges

Because it links every executed process back to a specific parent process and logon session, Event ID 4688 is foundational for building process trees, detecting process injection or spawning anomalies, and reconstructing an attacker's or user's actions on an endpoint. It is one of the core telemetry sources SOC analysts rely on for behavioral detection (e.g., an Office application spawning *powershell.exe* or *cmd.exe*.

6. Audit Configuration

To generate Event ID 4688, **Process Creation** auditing must be explicitly enabled — it is not logged by default on most Windows systems.

Steps used to enable auditing:

1. Open Administrator Command Prompt
2. Check Current Audit Configuration

```
auditpol /get /subcategory:"Process Creation"
```

3. Enable Audit Process Creation for Success
4. Verified via command line using:

```
auditpol /get /subcategory:"Process Creation"
```

To capture the **command line** used to launch each process (not just the process name), an additional policy was enabled:

```
C:\Windows\system32>auditpol /get /subcategory:"Process Creation"
System audit policy
Category/Subcategory      Setting
Detailed Tracking
  Process Creation        Success
```

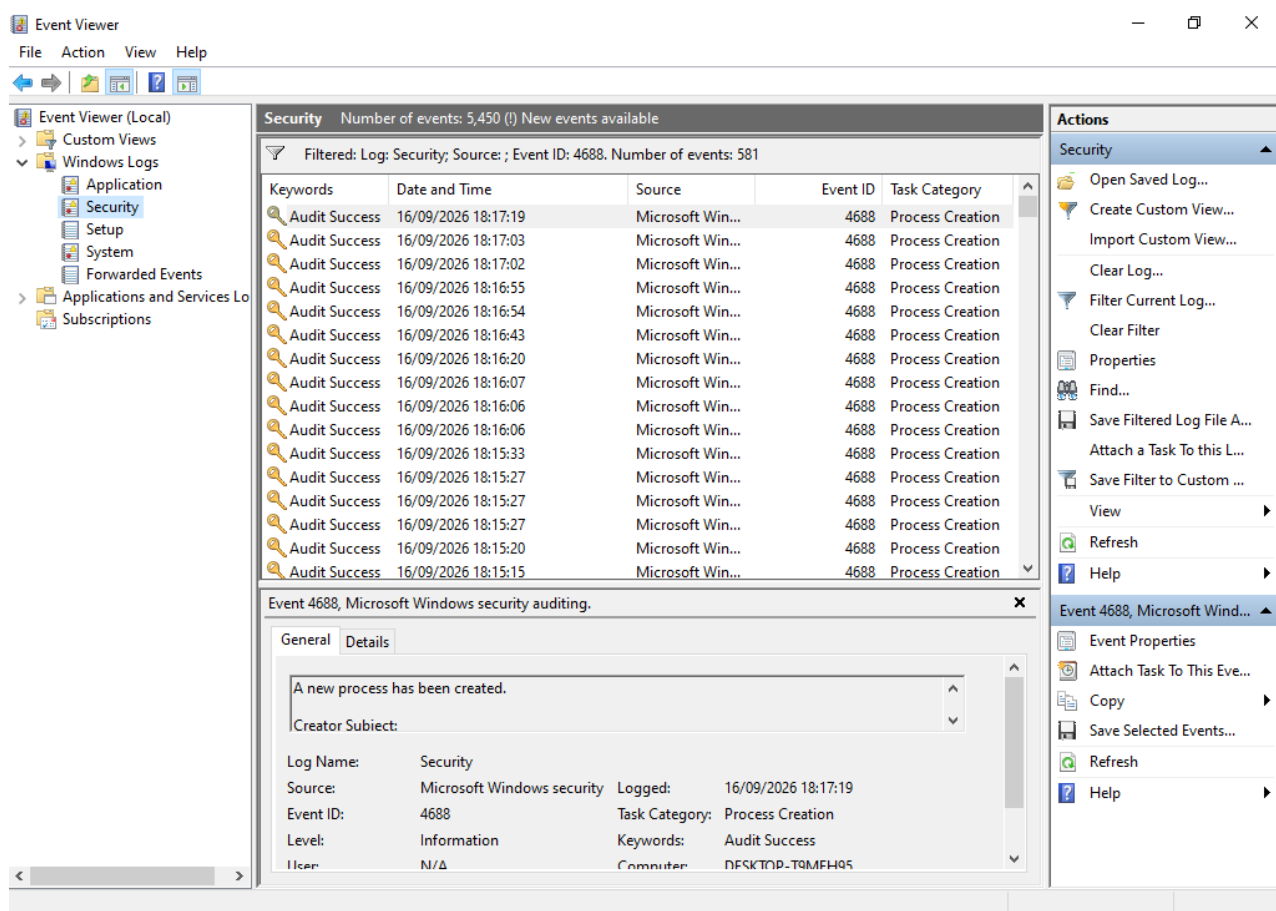
```
C:\Windows\system32>reg query "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v ProcessCreationI
ncludeCmdLine_Enabled

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit
  ProcessCreationIncludeCmdLine_Enabled    REG_DWORD    0x1
```

7. Methodology

The investigation followed this workflow:

- Enabled Process Creation auditing (and command-line auditing) via Local Group Policy
- Generated normal, controlled process activity on the VM to produce representative 4688 events
- Opened Event Viewer and navigated to Security Logs.
- Applied a custom filter for Event ID 4688
- Selected relevant events for analysis based on timing and process relevance
- Reviewed the General and Details (XML) tabs of each event to extract structured field data
- Cross-referenced the Subject Logon ID from each 4688 event against the corresponding Event ID 4624 logon event to confirm session context.



8.Event ID 4688 Evidence Analysis

Event ID	Timestamp	Subject -Logon ID	New Process	New Process ID	Creator Process Name	Creator Process ID	Command Line
4688	13:41:24	0x48780	TaskHost.exe	0x1d24	Svchost.exe	0x394	C:\\Windows\\system32\\backgroundTaskHost.exe
4688	13:41:44	0x49780	UserOOBEBroker.exe	0x1fb0	Svchost.exe	0x394	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe

GeneralDetails

Process Information:

New Process ID:

0x1d24

New Process Name:

C:\\Windows\\System32\\backgroundTaskHost.exe

Token Elevation Type:

%%1938

Mandatory Label:

Mandatory Label\\Low Mandatory Level

Creator Process ID:

0x394

Creator Process Name:

C:\\Windows\\System32\\svchost.exe

Log Name:

Security

Source:

Microsoft Windows security

Event ID:

4688

Level:

Information

User:

N/A

OpCode:

Info

More Information:

Event Log Online Help

Logged:

21/09/2026 13:41:24

Task Category:

Process Creation

Keywords:

Audit Success

Computer:

DESKTOP-T9MEH95

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Close

GeneralDetails

Process Information:

New Process ID:

0x1fb0

New Process Name:

C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe

Token Elevation Type:

%%1938

Mandatory Label:

Mandatory Label\\Medium Mandatory Level

Creator Process ID:

0x394

Creator Process Name:

C:\\Windows\\System32\\svchost.exe

Log Name:

Security

Source:

Microsoft Windows security

Event ID:

4688

Level:

Information

User:

N/A

OpCode:

Info

More Information:

Event Log Online Help

Logged:

21/09/2026 13:41:44

Task Category:

Process Creation

Keywords:

Audit Success

Computer:

DESKTOP-T9MEH95

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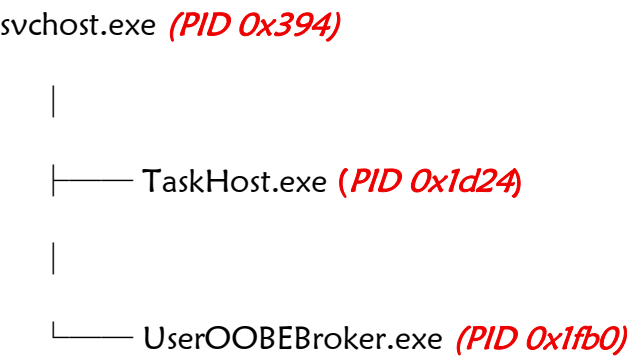
Close

9. Process / Parent-Child Analysis

The first event shows svchost.exe creating TaskHost.exe at 13:41:24. The creator process ID is 0x394 and the newly created process ID is 0x1D24.

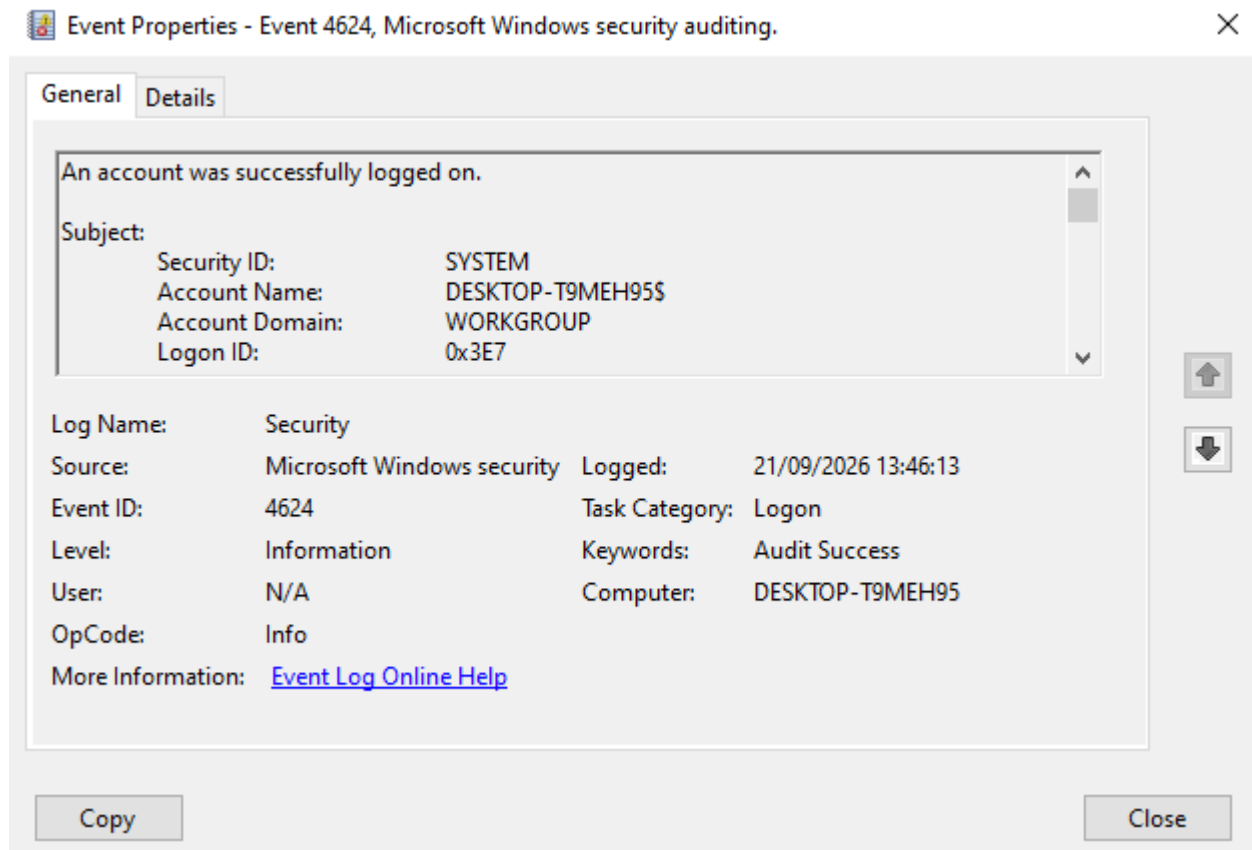
The second event shows svchost.exe creating UserOOBEBroker.exe at 13:41:44. It has the same creator process ID (0x394) as the previous svchost.exe event, indicating that both recorded child processes were associated with the same creator-process instance.

- Process Tree



10. Event 4624 + 4688 Correlation

Field	Event 4624 (Logon)	Event 4688 (Process Creation)
Timestamp	13:46:13	13:41:24
User	N/A	N/A
Logon ID	0x3E7	0x48780
Activity	Logon	Process execution



11. Timeline

Time	Event ID	User	Process J Activity	Observation
13:46:13	4624	N/A	Logon	Successful logon event recorded after the analyzed 4688 events.
13:41:24	4688	N/A	Process Execution	Created by svchost.exe

12. Findings

- a. Event ID 4688 successfully recorded process creation activity.
- b. Parent-child relationships were identified, including svchost.exe → TaskHost.exe/UserOOBEBroker.exe.
- c. Creator Process IDs and timestamps provided additional context for reconstructing process activity.
- d. Command-line information was available for the analyzed 4688 events.
- e. The collected activity represents **normal process creation in the controlled Windows VM**; no malicious activity was established from this evidence.

13. Limitations

- The investigation was conducted in a controlled Windows VM environment, which limits the diversity and volume of process activity compared to a production endpoint.
- Only normal, generated process activity was available for analysis; no adversarial or malicious activity was introduced.
- Event ID 4688 alone does not provide complete endpoint visibility — it does not capture network activity, file writes, or in-memory behavior, and should be correlated with other event sources (e.g., Sysmon Event ID 1) for deeper analysis.
- Command-line information in 4688 events is only available if the "Include command line in process creation events" policy is explicitly enabled; without it, command-line arguments are not logged.

14. Conclusion

This investigation demonstrated how Windows Event ID 4688 provides granular, timestamped visibility into process execution on an endpoint, including the identity of the process, its parent, and — where enabled — the exact command used to launch it. By correlating 4688 events with Event ID 4624 logon records, it was possible to tie process activity back to a specific authenticated session, reinforcing the value of process creation auditing as a core data source for reconstructing user and system activity in DFIR and SOC investigations.

15. MITRE ATT&CIX Relevance

Technique ID	Technique Name	Relevance
T1059	Command and Scripting Interpreter	Applicable if command-line/interpreter execution (e.g. cmd.exe, powershell.exe) is observed in evidence.
T1036	Masquerading	Applicable only if a process name/path inconsistency is observed in evidence

Note: Mapping should not be forced — The technique is clearly supported by the evidence, this section can state that no ATT&CK technique was conclusively observed.

16. References

- Microsoft Learn — 4688(S): A new process has been created.
- Microsoft Learn — Audit Process Creation
- Microsoft Learn — Command line process auditing
- MITRE ATT&CK — <https://attack.mitre.org/>

